

Sentiment Analyzer — Technical Report

A Local NLP Pipeline with RAG-Based Explanation

Course: Natural Language Processing

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1. Problem Description

In the beginning I started thinking about what problem to solve for this project. I thought of sentiment analysis tools but they tell you what the sentiment is and they never explain why. I also thought that if a user gets a "Very Negative" label they don't know what triggered it. So I decided to change this problem to make a system that reads a review and explains its reasoning.

I also wanted it to work in English and Spanish because I use both of them and I thought it would be a good challenge.

2. System Design and Workflow

I designed the system in five steps. In the beginning I thought of writing a single script but I decided to split the code into six independent modules because it makes it easier to test.

The first thing I did was clean the input. I remove URLs and special characters and I detect the language automatically. Then I built the classifier just how it was shown on HuggingFace and pass the clean text to a multilingual BERT model. It outputs the probability for five levels. I looked it up and I choose to use confidence scores for all categories because it's more informative.

After this I convert the review into an embedding vector to capture the meaning. Then I used cosine similarity to retrieve the two most similar past reviews. With this I build a prompt for Mistral 7B. I did this because we want the model to use the RAG component. I chose to put the temperature very low (0.3) because otherwise it was too creative. Finally I add the review to the memory.

3. Model Selection and Justification

In this case I searched online to choose the models because it affects the quality and if my laptop can run it.

For the sentiment I found out about [nlptown/bert-base-multilingual-uncased-sentiment](#) and I tried it. I used this because it is fine-tuned on product reviews. I didn't use a generative LLM because classifiers are faster.

For embeddings I used [paraphrase-multilingual-MiniLM-L12-v2](#). I tried other alternatives but this one worked correctly and it handles English and Spanish well.

For the LLM I chose Mistral 7B using Ollama. I thought of using LLaMA-70B but it's so big of a model that my laptop couldn't run it. I also tried OPT-125M but it didn't work very well because it couldn't produce good explanations. Mistral 7B finally worked and it didn't take too long.

4. Implementation Details

I built the project using Python and libraries like transformers, scikit-learn and gradio.

In this case I also used lazy model loading. I did this because otherwise it was taking too long to start. The model only loads when I use it. For the vector store I built a simple in-memory store using numpy arrays because using ChromaDB was taking too much configuration.

The program failed several times with the prompt, but I fixed it and finally it worked. One example of this error happened because I didn't put the right examples. Then I used two fixed examples to establish the format and then the retrieved reviews. For the interface I used Gradio Blocks because I wanted to show all the steps with pictures.

5. Results, Limitations and Possible Improvements

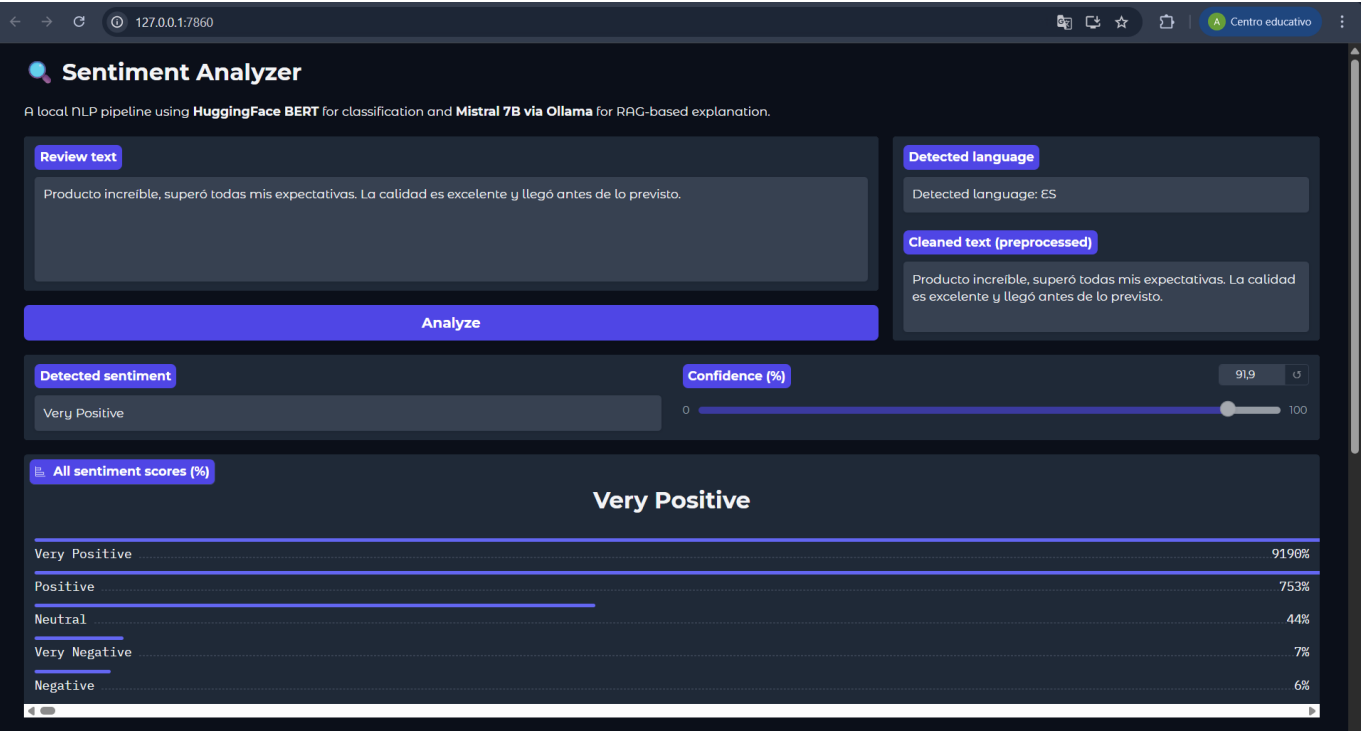
I tested it with English and Spanish reviews and it worked well. As you can see the RAG component makes the explanations much better after you put five or six reviews.

But it didn't work very well for everything because I think there are some limitations. The biggest problem is that the memory is deleted when you restart the app. The program also failed sometimes because the BERT model truncates at 512 characters, so if the review is too long it couldn't perform well at all.

As I said I think I would need to change the memory to ChromaDB so the data is not lost. I also think I would add a confidence threshold so it doesn't give a bad label if it's not sure. Now I would paste pictures of the interface with their results:

6. Screenshots

Very Positive review



Similar reviews retrieved from history:

1. (similarity: 75.29%) → Very Positive

This product completely exceeded my expectations. The build quality is outstanding and it arrived two days early. I have...

2. (similarity: 17.66%) → Very Negative

Total waste of money. It broke after three days and customer service ignored my emails completely. Avoid at all costs....

LLM Explanation (Mistral + RAG)

El usuario destaca que el producto excedió sus expectativas y la calidad es excelente. La palabra "increíble" transmite un alto nivel de satisfacción, mientras que "superó todas mis expectativas" indica una gran impresión positiva. Además, la frase "llegó antes de lo previsto" sugiere un servicio rápido y eficiente.

Session history

Refresh history

Review (preview)	Sentiment	Confidence
This product completely exceeded my expectations. The build ...	Very Positive	94.3%
Total waste of money. It broke after three days and customer...	Very Negative	96.98%

Usar vía API · Construido con Gradio · Configuración

Very Negative review

Sentiment Analyzer

A local NLP pipeline using HuggingFace BERT for classification and Mistral 7B via Ollama for RAG-based explanation.

Review text

Worst purchase I have ever made. The description was misleading, the quality is terrible and it does not work as advertised.

Detected language

Detected language: EN

Cleaned text (preprocessed)

Worst purchase I have ever made. The description was misleading, the quality is terrible and it does not work as advertised.

Analyze

Detected sentiment

Very Negative

Confidence (%)

96.41

All sentiment scores (%)

Very Negative

Very Negative	96.41%
Negative	33.4%
Neutral	19%
Very Positive	4%
Positive	2%

Similar reviews retrieved from history:

1. (similarity: 45.5%) + Very Positive

This product completely exceeded my expectations. The build quality is outstanding and it arrived two days early. I have...

2. (similarity: 40.07%) + Very Positive

Producto increíble, superó todas mis expectativas. La calidad es excelente y llegó antes de lo previsto...

LLM Explanation (Mistral + RAG)

The reviewer expresses strong dissatisfaction with the product due to several reasons: a misleading description, poor quality, and failure to function as promised. These negative aspects convey frustration and regret, leading to a very negative sentiment.

Session history

Refresh history

Review (preview)	Sentiment	Confidence
This product completely exceeded my expectations. The build ...	Very Positive	94.3%
Total waste of money. It broke after three days and customer...	Very Negative	96.98%

Usar via API · Construido con Gradio · Configuración

Neutral review

Sentiment Analyzer

A local NLP pipeline using HuggingFace BERT for classification and Mistral 7B via Ollama for RAG-based explanation.

Review text

It works fine but nothing special. Does the job, no complaints but no excitement either.

Detected language

Detected language: EN

Cleaned text (preprocessed)

It works fine but nothing special. Does the job, no complaints but no excitement either.

Analyze

Detected sentiment

Confidence (%)

Neutral

0100

All sentiment scores (%)

Neutral

Neutral	5843%
Positive	3701%
Negative	246%
Very Positive	184%
Very Negative	25%

Similar reviews retrieved from history:

1. (similarity: 29.46%) → Very Positive

Producto increíble, superó todas mis expectativas. La calidad es excelente y llegó antes de lo previsto...

2. (similarity: 26.8%) → Very Positive

This product completely exceeded my expectations. The build quality is outstanding and it arrived two days early. I have...

LLM Explanation (Mistral + RAG)

The reviewer describes the product as functioning adequately without any significant issues or negative comments, but also mentions a lack of excitement or exceptional qualities. This indicates a neutral sentiment since the product is meeting expectations without exceeding them. Phrases like "fine" "does the job" and "no complaints but no excitement either" contribute to this assessment.

Session history

Refresh history

Review (preview)	Sentiment	Confidence
This product completely exceeded my expectations. The build ...	Very Positive	94.3%
Total waste of money. It broke after three days and customer...	Very Negative	96.98%

Usar via API · Construido con Gradio · Configuración

Positive review

Sentiment Analyzer

A local NLP pipeline using HuggingFace BERT for classification and Mistral 7B via Ollama for RAG-based explanation.

Review text

I am happy with this purchase. Nothing revolutionary but solid quality and good value for money.

Detected language

Detected language: EN

Cleaned text (preprocessed)

I am happy with this purchase. Nothing revolutionary but solid quality and good value for money.

Analyze

Detected sentiment

Positive

Confidence (%)

48.52

All sentiment scores (%)

Positive

Positive	4852%
Very Positive	4631%
Neutral	480%
Negative	22%
Very Negative	14%

Similar reviews retrieved from history:

1. (similarity: 55.32%) + Very Positive

This product completely exceeded my expectations. The build quality is outstanding and it arrived two days early. I have...

2. (similarity: 51.89%) + Very Positive

Producto increíble, superó todas mis expectativas. La calidad es excelente y llegó antes de lo previsto...

LLM Explanation (Mistral + RAG)

The reviewer expresses satisfaction with the purchase, indicating a positive sentiment. Phrases like "happy" and "good value for money" suggest that they are pleased with their decision to buy the product, even though it may not be revolutionary, the mention of solid quality further reinforces this positive sentiment.

Session history

Refresh history

Review (preview)

Sentiment

Confidence

This product completely exceeded my expectations. The build ...

Very Positive

94.3%

Total waste of money. It broke after three days and customer...

Very Negative

96.98%

Usar via API · Construido con Gradio · Configuración

Negative review

Sentiment Analyzer

A local NLP pipeline using HuggingFace BERT for classification and Mistral 7B via Ollama for RAG-based explanation.

Review text

La entrega se retrasó y el embalaje llegó dañado. El producto en sí es, en el mejor de los casos, mediocre.

Detected language

Detected language: ES

Cleaned text (preprocessed)

La entrega se retrasó y el embalaje llegó dañado. El producto en sí es, en el mejor de los casos, mediocre.

Analyze

Detected sentiment

Confidence (%)

Negative

0

59.87

100

All sentiment scores (%)

Negative

Negative

5987%

Very Negative

1994%

Neutral

1948%

Positive

63%

Very Positive

8%

Similar reviews retrieved from history:

1. (similarity: 53.33%) + Very Negative

Worst purchase I have ever made. The description was misleading, the quality is terrible and it does not work as adverti...

2. (similarity: 37.73%) + Very Positive

This product completely exceeded my expectations. The build quality is outstanding and it arrived two days early. I have...

LLM Explanation (Mistral + RAG)

La entrega retrasada y el embalaje dañado indican una frustración con la experiencia del usuario. Además, la calificación del producto como "mediocre" transmite un sentimiento negativo.

Session history

Refresh history

Review (preview)	Sentiment	Confidence
This product completely exceeded my expectations. The build ...	Very Positive	94.3%
Total waste of money. It broke after three days and customer...	Very Negative	96.98%